

Article

BinoForce: A Force-Based 3D Dynamic Label Layout Method Under Binocular Viewpoints [†]

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[†] This article is a revised and expanded version of a poster entitled ‘3D Label Layout with Binocular Optimization’, which was presented at the 2025 ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games, Jersey City, NJ, USA, 7–9 May 2025. The current version contains substantially more detailed experiments and analysis compared to the conference version.

Abstract: In 3D virtual environments, effective information presentation relies on clear label layouts to annotate complex objects. As label density increases, structured and adaptive layouts become critical to prevent visual clutter and maintain clarity. However, existing label layout methods often overlook the issue of double vision between labels caused by binocular disparity. Additionally, discrete update strategies in 3D label layouts struggle to maintain layout quality during viewpoint changes due to their limited real-time adaptability. To address these issues, this paper presents a binocular 3D dynamic label layout method based on a continuous updating strategy within a force-based framework, generating real-time binocular layouts to minimize visual confusion. Computational evaluation and a user study demonstrate that the proposed method significantly reduces double vision, as quantified by a reduction in double vision degree of up to 15.74%, along with label overlap and leader line crossings. The results further indicate that our approach surpasses existing methods in clarity, aesthetics, adaptability, and user satisfaction, while reducing label reading time and alleviating users’ physical and effort demands, particularly in high-density label environments.

Keywords: 3D label layout; virtual reality; information visualization; user studies



Academic Editors: Juan Ernesto Solanes Galbis and Jean-Baptiste Barreau

Received: 15 April 2025

Revised: 7 May 2025

Accepted: 13 May 2025

Published: 29 May 2025

Citation: Zheng, Z.; He, Y.; Yu, G.; Xu, X. BinoForce: A Force-Based 3D Dynamic Label Layout Method Under Binocular Viewpoints. *Electronics* **2025**, *14*, 2199. <https://doi.org/10.3390/electronics14112199>

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1. Introduction

Clear information presentation is critical for users in 3D environments, particularly in immersive virtual reality (VR) or augmented reality (AR) applications. Labeling complex 3D structures or compositions allows users to quickly understand the functionality or attributes of objects. A common approach to improving visual clarity is through textual annotations, which provide additional details about objects in the scene. As shown in Figure 1, these labels typically consist of three key components: the text label itself, an anchor point marking the specific object being described, and a connecting line, commonly referred to as a “leader line”. An effective label layout enhances interaction efficiency by providing key information at a glance, reducing the time and effort needed to search for relevant details. However, despite ongoing advancements, existing methods still face significant challenges when dealing with label layout in dynamic 3D scenes with changing viewpoints.

A critical yet largely unaddressed challenge in 3D label layouts is the detrimental impact of binocular double vision—both between labels themselves and between labels and scene objects—on user experience. Current label placement methods primarily eliminate overlaps in monocular optimization. While this achieves label separation in the central view (Figure 2e), it inevitably causes residual overlap in the right-eye view (Figure 2f) due to ignored binocular disparity. This phenomenon mirrors how holding a finger close to your eyes makes it appear to “jump” sideways when alternating between left-/right-eye views. Recent work has begun exploring binocular vision for label optimization. Peterson et al. [1] improved label clarity and readability by distributing overlapping labels at different depths, while Emsenhuber et al. [2] optimized label layouts under binocular viewpoints to prevent double vision caused by differing backgrounds. However, these approaches still neglect label–label and label–object double vision. To the best of our knowledge, no prior work has addressed the negative effects of binocular double vision on user experience within the specific context of 3D label placement.

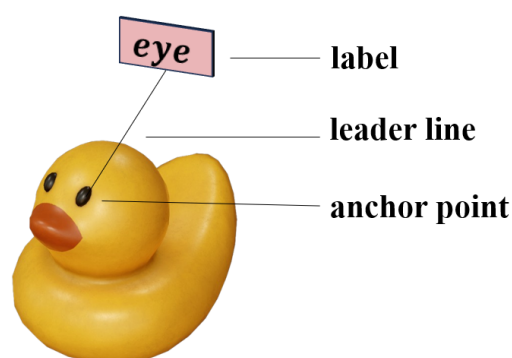


Figure 1. A 3D model with a label illustrating the typical labeling structure. It consists of three key components: the label itself, an anchor point on the object, and a leader line connecting the label to the anchor point.

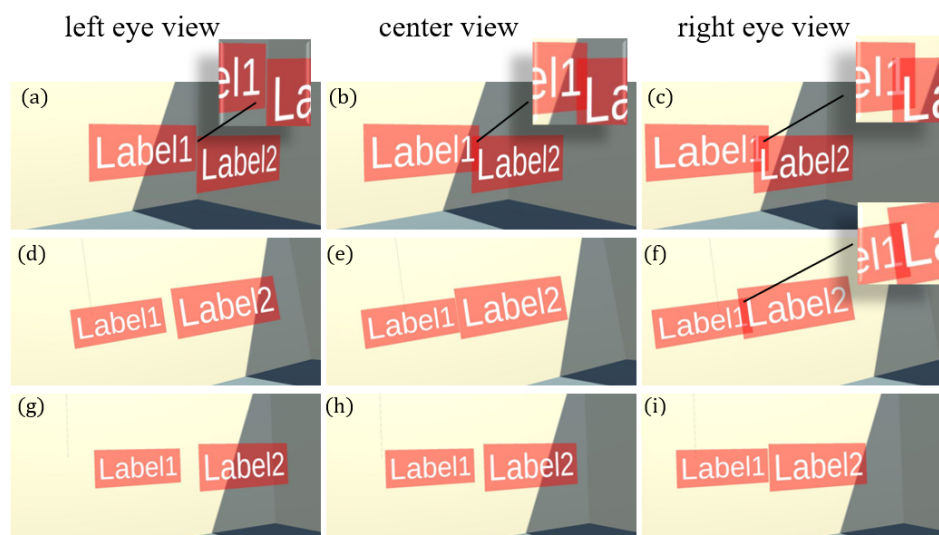


Figure 2. The results of handling label overlap under different perspective conditions are presented. Three virtual cameras are placed at the left eye, right eye, and the midpoint between the two eyes to simulate the views from each perspective. Subfigures (a–c) show a comparison of the same overlapping label layout as observed from the left-eye, center, and right-eye perspectives. In subfigures (d–f), the results of label layout optimization from a monocular perspective are displayed. Although the overlap issue is resolved in the center view, the labels still overlap in the right-eye view. In contrast, subfigures (g–i) present the label layout optimization results from binocular viewpoints. These results show no overlap across all perspectives.

Another challenge in dynamic 3D label layout lies in maintaining the efficacy of discrete update strategies when the number of labels in the scene increases. While label layout optimization under variable user viewpoints has garnered attention [3–5], many image-space methods exhibit unpredictable layout changes with viewpoint shifts, diminishing consistency and user experience. To tackle this, Tatzgern et al. [6] proposed a view management method for placing external labels in 3D object space, representing a significant advancement in the labeling of static 3D objects. While Tatzgern et al. employed a discrete update strategy, it still suffers from suboptimal layouts at certain moments due to its discrete nature. Madsen et al. [7] further demonstrated that a discrete update strategy, calculated from a single initial viewpoint, generally outperforms continuous strategies. However, Madsen et al.'s approach, for instance, addressed label overlap primarily through parallax effects between label planes, inadvertently introducing double vision and reducing user comfort. More importantly, as label density increases, accurately positioning these planes to eliminate overlap across numerous viewpoints becomes exponentially more complex. This raises a critical question: can discrete update strategies remain advantageous when a high label count requires intricate and potentially conflicting arrangements to prevent overlap across multiple viewpoints? Therefore, when the number of labels in the scene increases, these limitations present a significant hurdle to realizing the full potential of discrete update strategies in dynamic 3D label layouts.

To address these challenges, our work has two primary goals: first, to reduce the double vision effect between labels and between labels and objects and improve user experience; and second, to alleviate the limitations of discrete update strategies when the number of labels increases. In this paper, we present a 3D dynamic label layout method based on a continuous updating strategy [8]. Our method simultaneously optimizes label positions for both left- and right-eye viewpoints, effectively reducing double vision between labels and between labels and objects caused by binocular disparity. To achieve smooth and adaptive label movement, we adopt a force-based placement mechanism, where labels dynamically adjust their positions under the influence of calculated forces in each frame. As double vision between labels caused by binocular disparity has rarely been explored in the existing literature, we conducted a user study to investigate its impact on users.

In the computational assessment, our method was evaluated in three scenarios, effectively mitigating double vision and outperforming baseline methods in reducing label overlap and leader line crossings. We further conducted a user study where participants completed a label-reading task while their task completion time, workload scores (NASA-TLX [9]), and subjective questionnaire were collected. The results indicate that, compared to the baseline methods, our approach enables users to read the label more quickly, particularly in high-density label environments. Moreover, it achieved significantly better performance in workload scores and subjective questionnaire responses.

In summary, our main contributions are as follows:

- We demonstrate that double vision between labels caused by binocular disparity negatively impacts user experience, introducing a binocular optimization strategy to address this issue under dynamic viewpoints.
- We propose a 3D dynamic label layout method based on a continuous updating strategy, enabling labels to update their positions under the influence of multiple forces.
- We evaluate the proposed method through computational assessment and a user study, demonstrating its overall superiority over baseline methods.

2. Related Work

The study of label layout can be traced back to the 1970s. In 1975, Imhof et al. [10] provided foundational guidelines for placing names on maps to improve both readability

and aesthetic quality. In recent years, automatic label layout has become an increasingly active research area. However, many label layout problems have been proven to be NP-hard [11], which means it is computationally infeasible to find exact solutions within polynomial time. As a result, researchers commonly rely on heuristic or approximation algorithms to address these challenges rather than seeking exact polynomial-time solutions.

Existing label layout methods can generally be classified into two categories based on the space in which labels are placed: image space (2D) and world space (3D). In image space, elements are positioned on the screen for readability and clarity (Section 2.1). In world space, labels adjust dynamically based on the viewer's distance and viewpoint, providing a more immersive and context-aware user experience (Section 2.2). The respective advantages and challenges of 2D and 3D label layouts will be presented in Section 2.3. For a more comprehensive review of label layout methods, we recommend referring to [11,12].

2.1. Label Layouts in Image Space

Hartmann et al. [13] introduced an image-space floating label system using dynamic potential fields. Although this system was effective, it did not address leader line crossings or maintain label coherence in dynamic scenes. They later implemented real-time label arrangements by projecting 3D scenes onto 2D space [4].

Image-driven methods are frequently used for 2D boundary labeling. Bekos et al. [14] proposed placing labels along polygonal boundaries, connecting them with non-overlapping leader lines. Their subsequent work [15] explored variations to improve aesthetic label placement. Heinsohn et al. [16] developed a boundary labeling system for 2D biological networks, offering four distinct label arrangement strategies and allowing dynamic focus region movement.

Deep learning has also been applied to 2D label placement. Qu et al. [17] proposed a Label Placement Graph Transformer (LPGT), formulating label placement as a node prediction problem on a graph. Their method uses attention mechanisms to capture label interactions and outperforms existing approaches on the AMIL dataset.

In mobile mixed reality, Shibata et al. [18] proposed an image-space labeling method tailored for mobile devices. Čmolič et al. [5] introduced a real-time labeling technique for illustrations, combining fuzzy logic with greedy optimization. Despite dealing with 3D objects, both methods relied heavily on image-space view management. Ekman [19] presented a force-based method for real-time labeling of moving objects on maps, which He et al. [20] later extended with a hybrid optimization approach to achieve smoother dynamic label movements.

Image-space methods are also extensively used in augmented reality. Tatzgern et al. [21] utilized hierarchical clustering to build a multi-level structure for efficiently managing large datasets within the user's field of view. Labels are dynamically selected based on user preferences and visual clutter. Similarly, Grasset et al. [3] proposed an image-driven approach that identifies significant regions within the image, applying geometric constraints derived from visual saliency detection and edge analysis.

2.2. Label Layouts in World Space

World-space label layouts offer significant advantages in enhancing users' spatial awareness and immersion. Tatzgern et al. [6] introduced a view management method arranging labels in 3D world space, addressing floating label issues, but its discrete updates can cause constraint violations with viewpoint changes. Sirk et al. [22] built on this, developing a label layout method for translucent scenarios (common in forensics) by constraining the search space to finite planes at varying depths, improving search efficiency. Pick et al. [23] presented an automated labeling system for immersive VR using shadow

volumes to detect occlusions and a force-based approach to reposition labels, though it did not address leader line crossings.

In augmented reality and mixed reality environments, world-space label layouts have also been widely explored. Zhou et al. [24] arranged labels alphabetically on multiple concentric circles surrounding 3D objects, allowing users to locate specific labels more efficiently. Building on this, Wei et al. [25] introduced a label-guided target localization method, further enhancing object identification efficiency in VR settings. Köpper et al. [26] proposed a comprehensive label layout strategy that integrates occlusion management, level-of-detail control, and temporal consistency, treating labels as integral 3D scene elements. More recently, Chen et al. [27] became the first to apply reinforcement learning to optimize moving object label layouts in AR environments, accounting for both the current and predicted future states of objects and labels to dynamically adjust layouts. Employing reinforcement learning, Gebhardt et al. [28] presented an approach to learn personalized cooperative labeling policies for mixed-reality systems, optimizing for minimal visual overload during visual search tasks by observing users' gaze patterns.

Furthermore, labels can be considered a form of user interface element. In recent years, the field of adaptive user interfaces (AUIs) has seen a surge in research. Adaptive user interfaces are defined as user interfaces that dynamically adjust their behavior and presentation based on changes in the user, environment, or task. Within MR environments, common approaches to implementing adaptive user interfaces include optimization-based methods [29–32], supervised learning methods [33], and reinforcement learning methods [27,34,35].

In the context of depth perception, binocular disparity provides users with an intuitive understanding of object positions within a scene. Peterson et al. [1] demonstrated that increasing the depth difference between overlapping labels could reduce visual search time by 24%. Subsequently, Peterson et al. [36] extended their research to far-field scenarios (100–120 m), revealing that separating overlapping labels in distant views significantly improved response times. Additionally, Emsenhuber et al. [2] optimized label layouts by rendering images from the user's binocular perspective, effectively utilizing scene information from both eyes. However, our experimental findings reveal that while binocular disparity enhances depth perception, it simultaneously introduces double vision, which degrades the overall visual experience.

2.3. Two-Dimensional vs. Three-Dimensional Label Layouts

While label placement in image space is generally simpler than in world space, it introduces inherent limitations. For instance, when users change their viewpoints, unpredictable layout changes can occur due to the transformation of projected 3D points. Tatzgern et al. [6] observed that label reordering often happens when users move through the environment. Moreover, image-space labels lack depth information, which can result in poor integration between labels and the 3D environment, ultimately reducing the quality of user interaction.

Three-dimensional label layouts, by contrast, offer the potential for more intuitive and spatially consistent labeling, better leveraging the user's understanding of the 3D environment. This is especially true for dynamic 3D label layouts, where labels can adapt and move within the 3D scene. However, this approach presents significant challenges. Algorithms must consider the complexity of the scene, the user's viewpoint, and the potential for label overlap [37]. Furthermore, ensuring label stability and preventing visual clutter becomes increasingly difficult as the number of labels increases [21]. Finally, adapting UI element layouts in real time to dynamic viewpoints while preserving spatial

relationships and minimizing cognitive load on the user remains a complex optimization problem [38].

To address the limitations of existing label layout methods in adapting to dynamic viewpoints and mitigating double vision caused by binocular disparity, this paper proposes a force-based real-time dynamic 3D label layout method under binocular viewpoints. This approach allows labels to move freely within 3D scenes, ensuring visual consistency and reducing occlusions for an enhanced user experience.

3. Preliminary Study

Binocular disparity arises from the slight difference in images captured by the left and right eyes due to their distinct viewpoints. In a 3D environment, each eye perceives a slightly offset view, and the brain integrates these images to construct a stereoscopic perception. Figure 2 illustrates label layouts from left-eye, central, and right-eye perspectives.

Binocular disparity can cause double vision between labels or between labels and objects. This occurs when depth differences result in the left and right eyes perceiving labels at misaligned positions, making visual fusion difficult. When two labels (or a label and an object) are positioned at significantly different depths, their disparity increases, preventing the visual system from merging them into a single coherent image. As a result, users may perceive duplicated or misaligned labels, leading to visual discomfort and reduced readability.

However, most existing layout methods optimize label placement only from a central viewpoint. As shown in Figure 2d–f, while overlap is resolved in the central view, it persists in the right-eye view. Consequently, when the brain combines both eye images into a unified 3D perception, users still experience double vision [39].

To investigate the impact of binocular parallax-induced double vision between labels on users, we conducted an exploratory user study. The evaluation considered readability, visual comfort, aesthetics, and interactivity of labels. In this work, the interactivity of labels refers to the ability of users to select a specific label within the 3D scene. User perceptions of this interactivity were qualitatively evaluated using Likert-scale questions, the specific design of which is detailed in the “Measures” Section of this chapter. This study was approved by our institution’s ethics committee.

Study design. We designed two scenes to compare the effects of double vision between labels. Scene 1 (Figure 3(left, b)): in the left-eye viewpoint, labels do not overlap, whereas in the right-eye viewpoint, labels overlap, resulting in double vision. Scene 2 (Figure 3(left, a)): labels do not overlap in either eye’s viewpoint, eliminating double vision. To ensure that the label overlap caused by binocular disparity was consistent across participants, we marked a fixed observation point in the scenes, requiring all users to stand at the same position. Additionally, label height was adjusted based on each participant’s height to align with their eye level. To mitigate potential order effects, we counterbalanced the order in which participants completed the two scenes. After observing both scenes, participants completed a Likert-scale questionnaire to assess readability, visual comfort, aesthetic appeal, and interactivity of the labels.

Participants. A total of 19 participants (11 female, 8 male) were recruited for this study. They range in the age from 20 to 45 ($M = 27.2$). Among them, 15 participants had prior experience using VR systems, while 4 had little to no experience. All participants had normal or corrected-to-normal vision and reported no significant prior history of motion sickness in VR environments. Informed consent was obtained from all participants before the study.

Study flow. In the study flow, participants were given a brief overview of the study's purpose. Participants observed layouts in both scenarios. In scene 1, double vision between labels occurs due to binocular disparity, whereas in scene 2, it is absent.

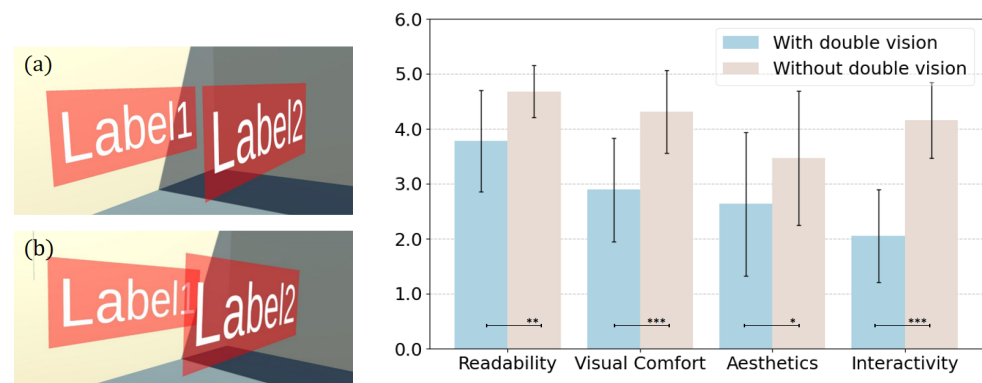


Figure 3. Scene of the preliminary study (left): (a) double vision is eliminated; (b) double vision occurs between labels. User evaluation results under the two experimental conditions (with and without double vision) are shown on the (right). Error bars indicate the 95% confidence intervals. ***, **, and * represent 0.1%, 1%, and 5% significance levels, respectively.

Measures. A five-point Likert scale was used to collect user feedback, with the following questions: Is the text on the labels easy to read (Q1, Readability)? Is reading the labels visually comfortable (Q2, Visual Comfort)? Does the label layout affect aesthetic appeal (Q3, Aesthetics)? In this context, are the speed and accuracy of selecting labels with a controller affected (Q4, Interactivity)?

Methods for data analysis. A paired-sample *t*-test was employed to assess whether significant differences exist in readability, visual comfort, aesthetic appeal, and interactivity. First, the Shapiro–Wilk test was conducted to assess whether our data approximately followed a normal distribution. Subsequently, a paired-sample *t*-test [40] was performed to evaluate the significance of the *p*-values. If significant differences were found, further analysis of effect sizes was conducted. Cohen's *d* [41] was used to evaluate the effect size of the differences, where values in the range of [0.5, 0.8) indicate a medium effect, and values greater than 0.8 signify a large effect.

Results. Prior to analysis, we conducted the Shapiro–Wilk test, confirming that the data approximately followed a normal distribution. As is shown in Figure 3(right), a paired-sample *t*-test showed significant differences between the two conditions across all measures. The experimental results (as shown in Table 1) indicate that the most significant differences between the two experimental conditions were observed for visual comfort and interactivity, followed by readability and aesthetics.

Table 1. Results of paired-sample *t*-tests for the preliminary study. ***, **, and * represent 0.1%, 1%, and 5% significance levels, respectively.

Dimension	<i>t</i>	<i>p</i>	Cohen's <i>d</i>
Readability	−3.545	0.002 **	0.813
Visual Comfort	−4.748	<0.001 ***	1.089
Aesthetics	−2.446	0.025 *	0.561
Interactivity	−4.595	<0.001 ***	1.054

Discussion. Compared to labels with double vision, those without double vision significantly enhance visual comfort and interactivity, while moderately improving readability and aesthetics. These results suggest that eliminating double vision is crucial for improving user experience.

4. Method

We propose a force-based continuous updating strategy designed to address the challenges of 3D dynamic label layouts under binocular viewpoints. This section elaborates on our method from two perspectives: binocular optimization (Section 4.1) and the force-based strategy (Section 4.2).

4.1. Optimization Under Binocular Viewpoints

The double vision between labels arises due to binocular disparity, as traditional layout optimization strategies rely on a single-camera viewpoint (as shown in Figure 4a). To address this issue, we optimize the layout from a binocular perspective by simulating the user's left- and right-eye views using two virtual cameras positioned on either side of the main camera (as shown in Figure 4b). Following MacLachlan et al.'s [42] study on the average interpupillary distance, we set the distance between the virtual cameras to a fixed value of 6.4 cm. In each viewpoint, scene information is processed independently, and the results are then jointly optimized to obtain the final outcome. For example, when calculating label overlap, we compute the overlapping area ratio between labels and use the larger result from the two viewpoints as the final measure, integrating it into the force-based layout update.

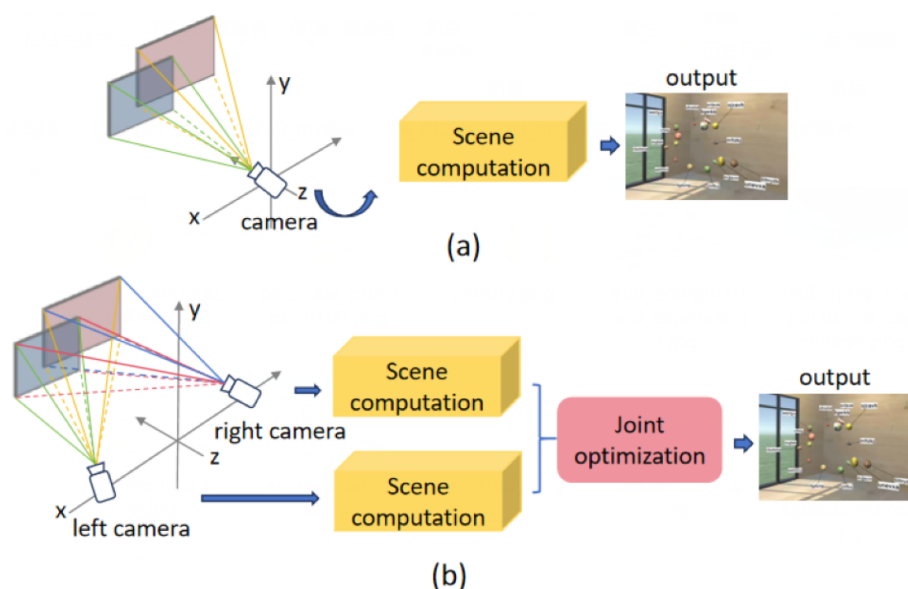


Figure 4. Monocular and binocular label layout optimization methods. The monoscopic optimization strategy computes the 3D scene from the perspective of the main camera only (a). In contrast, the binocular optimization strategy employs two virtual cameras to simulate the left- and right-eye perspectives, allowing the scene to be perceived and computed from both viewpoints (b). Both monocular and binocular optimization methods output optimized 3D label layouts—arranging labels in 3D space to ensure clarity and avoid occlusion. Monocular methods, relying on a single viewpoint, may produce double vision, while binocular methods leverage dual-view information to mitigate this issue.

Figure 2g–i illustrate the effectiveness of this approach in mitigating label overlap from a binocular perspective. The results show that no overlap occurs in either the left-eye or right-eye view, thereby eliminating potential ghosting when users perceive the scene binocularly. Given that similar double vision issues also occur between labels and objects, we further extend this binocular optimization strategy to address label–object overlap.

4.2. Force-Based Updates

Each label is initially positioned at its corresponding anchor point and then continuously updated under the influence of forces. Figure 5 shows the forces applied to labels.

Repulsive force. To ensure a more even distribution of labels in the space, each label experiences a repulsive force from all other labels. As shown in Figure 5a, $label_i$ experiences repulsive forces from $label_j$ and $label_k$. The repulsive force exerted on $label_i$ by $label_j$ is inversely related to the distance between them, which is calculated as follows:

$$F_{repulse}^i = \frac{1}{n-1} \sum_{j \neq i} \frac{1}{\|l_i - l_j\|} \cdot \vec{d}_{ij}$$

where

- $F_{repulse}^i$ is the repulsive force experienced by $label_i$.
- n is the total number of labels in the space.
- l_i represents the coordinates of $label_i$.
- l_j represents the coordinates of $label_j$.
- $\|l_i - l_j\|$ is the Euclidean distance between $label_i$ and $label_j$.
- $\vec{d}_{ij}$ is the unit vector pointing from $label_j$ to $label_i$.

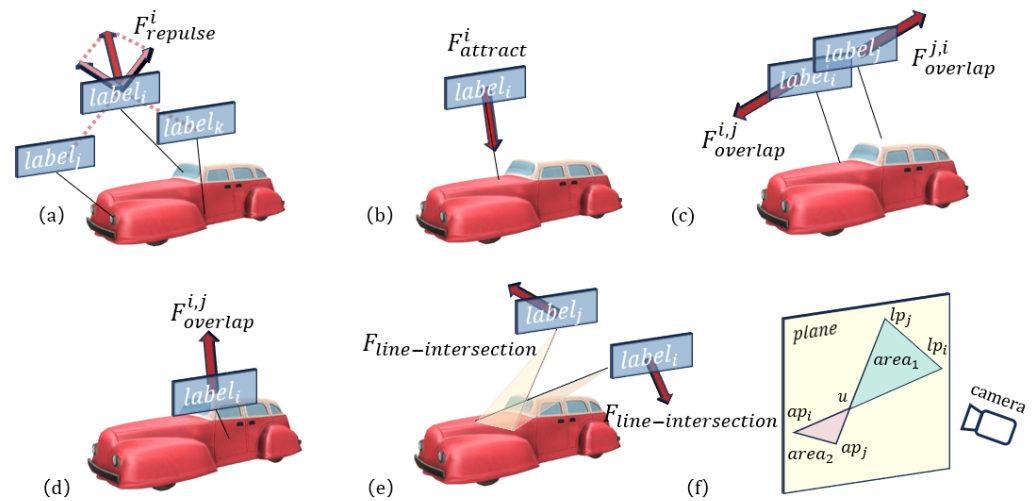


Figure 5. Illustration of the five forces applied to labels. (a) The repulsive force exerted on a label by other labels; (b) the attractive force pulling a label towards its corresponding anchor point; (c) the repulsive force between overlapping labels; (d) the repulsive force on a label due to overlap with an object; (e) the force introduced to resolve leader line crossings; (f) the method for calculating the magnitude of the force resolving leader line crossings.

Attractive force. To prevent labels from being too far from their corresponding anchor points, each label is subjected to an attractive force (Figure 5b) from its anchor point. The attractive force exerted on $label_i$ by its corresponding anchor point is positively correlated with the distance between them, which is calculated as follows:

$$F_{attract}^i = \|l_i - a_i\| \cdot \vec{d}_i$$

where

- $F_{attract}^i$ is the attractive force experienced by $label_i$.
- l_i represents the coordinates of $label_i$.
- a_i represents the coordinates of the anchor point corresponding to $label_i$.
- $\vec{d}_i$ is the unit vector pointing from l_i to a_i .

Overlapping force. If labels overlap with each other or with an object, a repulsive force (Figure 5c,d) is applied to minimize visual interference. Adapting Pick et al. [23], we calculate the overlap area and introduce the overlapping ratio—the proportion of a label's overlap to its total area—ensuring size-independent measurement. Then, we define $R_{i,j}$, the label overlapping ratio between $label_i$ and $label_j$, or $label_i$ and $object_j$, under binocular viewpoints, which is calculated as follows:

$$R_{i,j} = \max(R_{i,j}^{left}, R_{i,j}^{right})$$

where

- $R_{i,j}^{left}$ is the overlapping ratio from the viewpoint of the left eye.
- $R_{i,j}^{right}$ is the overlapping ratio from the viewpoint of the right eye.

The magnitude of the overlapping force $F_{overlap}^{i,j}$ is directly proportional to $R_{i,j}$, which is calculated as follows:

$$F_{overlap}^{i,j} = m \cdot R_{i,j} \cdot \vec{d}_{ij}$$

where

- m is a constant empirically set to 4 to enhance the effect of the overlapping force. This value was determined through a series of experiments where we varied m and observed the resulting label layouts. Values significantly below 4 caused insufficient label separation, while values much higher than 4 pushed labels too far apart, compromising visual coherence.
- $R_{i,j}$ is the label overlapping ratio between $label_i$ and $label_j$ (or $label_i$ and $object_j$).
- $\vec{d}_{ij}$ represents a unit direction vector pointing from $label_j$ to $label_i$ (or $object_j$ to $label_i$), projected onto the plane defined by the normal vector of the line connecting the user's position to the midpoint of the two labels (or a label and an object).

Leader line intersection force. Leader lines intersect when the user moves or changes viewpoint in a VR environment. To address this, a new force (Figure 5e) that acts on the labels associated with intersecting leader lines is introduced. The magnitude of the force is as follows:

$$\|F_{line-intersection}^{i,j}\| = D_{i,j} = \frac{\min(area_1, area_2)}{(area_1 + area_2)}$$

where

- $D_{i,j}$ represents the degree of leader line crossings between $label_i$ and $label_j$.
- $area_1$ denotes the area of the triangle formed by u , lp_i , and lp_j , where the intersecting leader lines are projected from world space into the main camera's screen space; the intersection point of the two lines is denoted as u , with lp_i and lp_j corresponding to $label_i$ and $label_j$, respectively (Figure 5f).
- $area_2$ corresponds to the area of the triangle formed by u , ap_i , and ap_j , where ap_i and ap_j represent their associated anchor points.

The force direction of $label_i$ is denoted as dir_i , which is calculated as follows:

$$\vec{dir}_i = \begin{cases} -\vec{n}_i, & \text{if } \theta_2 > \theta_1 \\ \vec{n}_i, & \text{otherwise} \end{cases}$$

where

- $\vec{dir}_i$ is the force direction of $label_i$.
- $\vec{n}_i$ is the normal vector of the plane defined by the position of anchor points a_i , a_j , and label l_i .

- $\vec{n}_j$ is the normal vector of the plane defined by the position of anchor points a_i , a_j , and label l_j .
 - θ_1 represents the angle between vectors $\vec{n}_i$ and $\vec{n}_j$.
 - θ_2 represents the angle between $\vec{n}_i$ and $-\vec{n}_j$.
- θ_1 and θ_2 are calculated as follows:

$$\theta_1 = \cos^{-1} \left(\frac{\vec{n}_i \cdot \vec{n}_j}{|\vec{n}_i| |\vec{n}_j|} \right)$$

$$\theta_2 = \cos^{-1} \left(\frac{\vec{n}_i \cdot (-\vec{n}_j)}{|\vec{n}_i| |-\vec{n}_j|} \right)$$

We introduce the force F_{circ} proposed by Pick et al. [23], aimed at generating a circular layout for the labels to enhance aesthetic appeal. After computing the aforementioned forces, we obtain the resultant force acting on $label_i$ by calculating the weighted sum of its individual force components, as shown in the equation below.

$$F_i = w_1 F_{repulse}^i + w_2 F_{attract}^i + w_3 \sum_{j \neq i} F_{overlap}^{i,j} + w_4 \sum_{k \neq i} F_{line-intersection}^{i,k} + w_5 F_{circ}^i \quad (1)$$

where j represents other labels and objects in the scene, while k refers specifically to other labels. The weights are empirically set as follows: $w_1 = 0.02$, $w_2 = 0.09$, $w_3 = 0.06$, $w_4 = 0.8$, and $w_5 = 0.03$.

In each frame, the resultant force F_i is computed for each label $label_i$ in each scene. The label's displacement x is then determined based on F_i , and its coordinate l_i is updated as follows:

$$l_i = l_i + x$$

To enhance stability, the computed displacement is further scaled by an empirical factor of 0.01. Additionally, to enhance readability for users on the move, we adjusted the label orientation in each frame so that it consistently faces the user.

5. Computational Assessment

5.1. Evaluating Double Vision in Binocular and Monocular Viewpoints

To assess the effectiveness of the binocular optimization strategy, we propose a novel metric, double vision degree (DBV), to compare scenarios with and without binocular optimization. DBV quantifies the proportion of double vision area to the total label area for each label, averaged over each time step. The double vision area is estimated as the absolute difference between the overlapping label area in the left-eye and right-eye viewpoints.

We conducted experiments in three different scenes: fruit, virus, and helicopter (Figure 6). The virus scene exemplifies the challenges of label placement at a small scale, demanding precise label localization. Conversely, the helicopter scene represents the challenges associated with labeling a large-scale object, requiring an effective approach for managing labels and clearly presenting its components. Finally, the fruit scene, comprising multiple independent models, is designed to evaluate our method's performance in multi-object scenes. For the label layout results, please refer to the supplementary materials. To simulate dynamic conditions, we set the camera to continuously move within the scene. It followed a circular trajectory parallel to the ground around the scene center. Every five seconds, the camera had a 50% probability of stopping for five seconds or continuing with linear motion in a random direction parallel to the ground for another five seconds.

Table 3. The comparison of three methods in terms of leader line crossings and label overlap.

Scene	OLR			LCD		
	Baseline	Hedgehog	BinoForce	Baseline	Hedgehog	BinoForce
Fruit	0.13079	0.08397	0.02005	0.00004	0.01998	0.00285
Virus	0.40784	0.32541	0.20177	0.00118	0.02048	0.01747
Helicopter	0.69942	0.32311	0.15847	0.00020	0.02748	0.01287

6. User Study

We conducted a user study to evaluate whether our method facilitates faster comprehension of scene information in dynamic environments. This study was approved by our institution's ethics committee.

6.1. Methods

Apparatus. Our label layout application was developed using the Unity game engine (version 2022.3.8f1c1), utilizing the XR Interaction Toolkit (version 2.4.3) to manage virtual reality interactions. The application was tested on a Nolo VR headset, which was connected to a laptop equipped with an Intel i9-13900 2.20 GHz CPU, an NVIDIA GeForce RTX 4090 GPU, and 32 GB of RAM.

Participants. We recruited 24 participants (13 female, 11 male) aged between 20 and 45 years ($M = 26.5$). Among them, 17 had prior experience with VR systems, while 7 had little to no experience. All participants had normal or corrected-to-normal vision and reported no significant history of motion sickness in VR environments. Informed consent was obtained from all participants prior to the study.

Study design. We employed a within-subject design with nine scenes, defined by two independent variables: METHOD (*Baseline*, *Hedgehog*, *BinoForce*) and DENSITY (*Low*, *Medium*, *High*). Each scene consisted of a $1\text{ m} \times 1\text{ m} \times 1\text{ m}$ cube, where 10, 17, and 24 objects were placed in the cube in the low-, medium-, and high-density conditions, respectively (Figure 7). Each object was associated with a label displaying a randomly generated six-letter English string, as is illustrated in Figure 8(left). Participants were asked to move freely within a range of 1 to 2 m from the cube, observing the generated label layout from any perspective. In each scene, participants performed a label-reading task: at random intervals, a randomly selected object was highlighted in blue. Upon seeing the highlight, the participant was required to read the corresponding label aloud. Each scene included seven repetitions of this task. In total, we collected 1512 trials (24 participants $\times$ 3 methods $\times$ 3 density levels $\times$ 7 repetitions) of data. A Latin Square design was employed to counterbalance the presentation order and mitigate learning and fatigue effects.

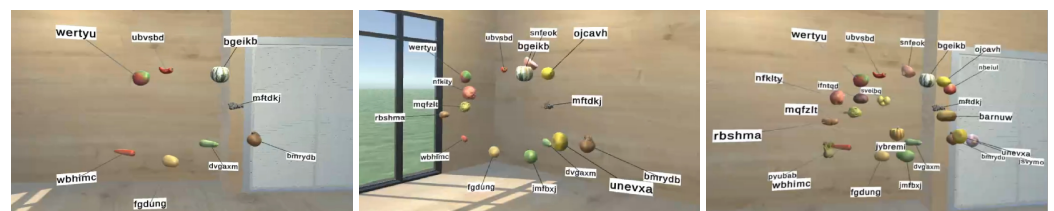


Figure 7. User study scenarios under three density conditions: the low-density scenario contains 10 labels (left), the medium-density scenario contains 17 labels (middle), and the high-density scenario contains 24 labels (right). All illustrated scenarios utilize our label layout method.

Study flow. Participants first completed an informed consent form and a demographic questionnaire, followed by a brief introduction to the study's purpose. Before the main experiment, they underwent a short training session in a simple virtual scene to familiarize

themselves with the equipment and tasks. Once they confirmed readiness, the experiment commenced. Each participant completed a total of 63 (3 methods $\times$ 3 density levels $\times$ 7 repetitions) label-reading trials. Participants were instructed to complete the task as quickly and accurately as possible.

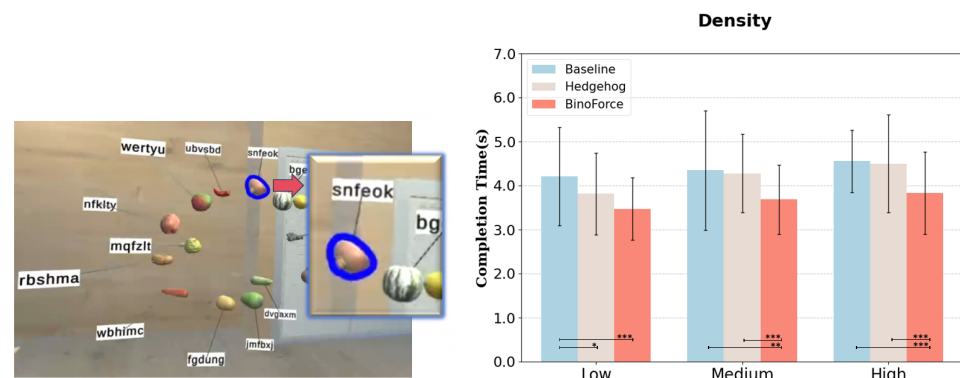


Figure 8. Example of the user study scenario (left) and task completion time across different label densities for the three methods (right). Error bars indicate the 95% confidence interval. ***, **, and * represent 0.1%, 1%, and 5% significance levels, respectively.

Measures. We recorded the task completion time (TCT), measured from the moment an object was highlighted until the participant read its label correctly. After the experiment, participants provided subjective feedback through two questionnaires: a standard NASA-TLX survey [9] to assess perceived workload and a five-point Likert scale evaluating the effectiveness of each method. For each method, users were required to answer the following five questions: Are the labels clear and easy to read? (Q1, Clarity) Is the label layout visually appealing? (Q2, Aesthetics) Does the label layout remain stable? (Q3, Stability) Does the algorithm effectively adjust the layout when quality deteriorates? (Q4, Adaptability) How satisfied are you with the overall layout? (Q5, Satisfaction) Finally, participants were asked to provide qualitative feedback on the strengths and weaknesses of each method.

Methods for data analysis. A one-way repeated measures ANOVA was conducted to compare task completion time across the three label density conditions. First, the Shapiro–Wilk test [40] was used to assess the normality of the data, which confirmed that our data followed a normal distribution. Next, we tested the sphericity assumption to evaluate the correlation among conditions. After confirming sphericity, we examined whether significant differences existed among the three methods. Finally, the Bonferroni method was applied for post hoc multiple comparisons to identify specific differences. To analyze the NASA-TLX survey and user questionnaire, the Friedman non-parametric test was used to determine whether significant differences existed among the methods for each metric. The Wilcoxon signed-rank test was then applied to assess the magnitude of these differences.

6.2. Results

Figure 8 presents the results of task completion times under varying label density conditions. For low density, METHOD was found to have a statistically significant effect on completion time, with a large effect size ($F_{2,46} = 34.898, p < 0.001$). For task completion time across methods, we found significant differences between *Baseline* and *Hedgehog* ($p < 0.05$), as well as between *Baseline* and *BinoForce* ($p < 0.001$). For medium density, METHOD was found to have a statistically significant effect on task completion time, with a large effect size ($F_{2,46} = 16.375, p < 0.001$). Our method took the least time to complete when compared to *Baseline* ($p < 0.01$) and *Hedgehog* ($p < 0.001$). The difference between *Baseline* and *Hedgehog* was not significant. For high density, METHOD was found to have a statistically significant

effect on completion time, with a large effect size ($F_{2,46} = 25.103, p < 0.001$). Our method took the least time to complete when compared to *Baseline* ($p < 0.001$) and *Hedgehog* ($p < 0.001$). The difference between *Baseline* and *Hedgehog* was not significant.

Figure 9 and Table 4 present the results of the NASA TLX survey. The Friedman test indicates that the method significantly affects mental demands, physical demands, temporal demands, and effort. However, there is no significant effect on performance and frustrations. In terms of mental demands, *Hedgehog* has a significant effect compared to *Baseline* ($Z = -2.818, p < 0.01$) and *BinoForce* ($Z = -3.707, p < 0.001$). For physical demands, *BinoForce* significantly affects both *Baseline* ($Z = -3.225, p = 0.001$) and *Hedgehog* ($Z = -3.659, p < 0.001$). Regarding temporal demands, our method shows a significant effect compared to *Baseline* ($Z = -3.216, p < 0.001$) and *Hedgehog* ($Z = -3.289, p < 0.001$). For effort demands, our method shows a significant effect compared to both *Baseline* ($Z = -2.708, p = 0.007$) and *Hedgehog* ($Z = -3.686, p < 0.001$).

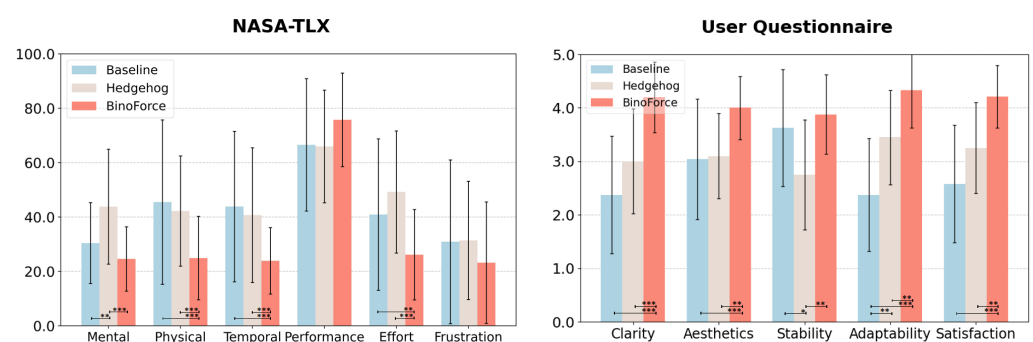


Figure 9. The results of the standard NASA-TLX survey (left) and the user questionnaire (right). Error bars indicate the 95% confidence interval. ***, **, and * represent 0.1%, 1%, and 5% significance levels, respectively.

Table 4. Friedman test results for the NASA-TLX survey (top) and user questionnaire (bottom).

Dimension	χ^2_2	p	W
Mental	15.929	<0.001	0.332
Physical	14.990	<0.001	0.300
Temporal	15.741	<0.001	0.328
Performance	2.141	0.343	0.045
Effort	8.458	0.015	0.176
Frustration	3.928	0.140	0.082
Dimension	χ^2_2	p	W
Clarity	24.376	<0.001	0.508
Aesthetics	16.088	<0.001	0.335
Stability	11.425	0.003	0.238
Adaptability	23.928	<0.001	0.498
Satisfaction	24.442	<0.001	0.509

Figure 9 and Table 4 shows the results of the user questionnaire. The Friedman test indicates that the method significantly affects clarity, aesthetics, stability, adaptability, and satisfaction. In terms of clarity, our method has a significant effect compared to *Baseline* ($Z = -3.957, p < 0.001$) and *Hedgehog* ($Z = -3.364, p < 0.001$). For aesthetics, *BinoForce* significantly affects both *Baseline* ($Z = -3.107, p = 0.002$) and *Hedgehog* ($Z = -3.520, p < 0.001$). Regarding stability, *Hedgehog* shows a significant effect compared to *Baseline* ($Z = -2.137, p = 0.033$) and *BinoForce* ($Z = -3.101, p = 0.002$). For adaptability, the data indicated significant differences between *Baseline* and *Hedgehog* ($Z = -2.671, p = 0.008$),

Baseline and *BinoForce* ($Z = -3.973, p < 0.001$), as well as *Hedgehog* and *BinoForce* ($Z = -2.979, p = 0.003$). For satisfaction, our method significantly affects both *Baseline* ($Z = -4.001, p < 0.001$) and *Hedgehog* ($Z = -3.079, p = 0.002$).

6.3. User Feedback

After completing the experiment, participants were asked to provide feedback on each method, highlighting its strengths and weaknesses.

For the *Baseline* method, several participants positively evaluated the method's stability, with representative comments including the following: "Highly stable and intuitive" (P9) and "The fixed relative positions between labels and objects allow rapid visual search once leader lines are established" (P10). One participant explicitly noted "This approach enhances label positional stability" (P20). Additionally, some participants (P6, P16) reported that the stability facilitated easier memorization of label locations. However, this stability also introduced occlusion issues, making label reading more challenging. For example, in terms of physical effort, some participants reported that "constantly adjusting the viewing angle to read labels was tiring" (P4) and "It took too much energy to read occluded labels" (P18). In terms of accuracy, one participant mentioned occasional misreading, such as mistaking "h" for "n" (P10).

Regarding the *Hedgehog* method, approximately half of the participants appreciated its adaptive nature, with comments such as the following: "The dynamic label adjustment effectively mitigates occlusion issues" (P17) and "Real-time label repositioning improves readability" (P5). However, several issues were raised: (1) Abrupt label movements. Some participants found that label positions changed too drastically, making it difficult to memorize their locations (P16, P19). Additionally, sudden updates during label reading forced participants to "constantly shift their gaze" (P12, P13), making it hard to keep up with the moving labels. (2) Unnecessary updates. Two participants mentioned that layout updates occasionally occurred even when not needed (P8, P11). (3) Occlusion. Since the layout was updated discretely rather than continuously, occlusion issues still occurred during movement (P1, P2, P11).

Compared to the other two methods, most participants found *BinoForce* to provide a clearer and more intuitive label layout, with representative comments including the following: "The labels update their positions smoothly in real time without disruptive jumps" (P12), "Adaptive label movement according to viewpoint changes significantly improves legibility" (P2), and "The system responds instantaneously to user viewpoint changes" (P18). Some participants mentioned that the real-time updates occasionally felt unstable and distracting (P1, P20). However, others considered these updates to be "within an acceptable range" (P4, P6, P11).

6.4. Discussion

Figure 8 shows that our method consistently outperforms *Baseline* in task completion time, achieving statistically significant improvements. This indicates that our approach provides a tangible benefit to users regardless of the complexity of the scene. While its performance is comparable to *Hedgehog* in low-density conditions, it enables faster label reading in medium- and high-density scenes, highlighting the advantage of our method in more complex environments. As label density increases, the advantages of *Hedgehog*'s discrete update strategy gradually diminish. This is likely because labels cannot update their positions in real time, leading to constraint violations during viewpoint changes—an issue that becomes more frequent with higher density. Therefore, we recommend adopting our continuous update strategy in such scenarios, especially when dealing with medium to

high label densities where real-time adaptation is crucial for maintaining label clarity and minimizing occlusion.

Figure 9 shows that, compared to the two baseline methods, our approach demonstrates lower physical demands, temporal demands, and effort. The statistically significant improvements in these categories suggest that our method is perceived as less physically taxing, allows users to complete tasks more quickly, and requires less overall exertion. This is a crucial finding, as reduced workload can lead to improved user satisfaction and reduced fatigue, particularly in demanding tasks. Regarding performance and frustration, no significant differences were observed among the three methods, suggesting that a larger-scale user study is needed to enhance statistical power and confirm significance. Additionally, user questionnaire results indicate that our method outperforms the baselines in terms of layout clarity, aesthetics, adaptability, and satisfaction. The statistically significant improvements across these categories suggest that users perceive our method as providing a more understandable, visually appealing, flexible, and ultimately more pleasing experience. These findings reflect the subjective preferences and overall impression of the users, which can significantly impact the adoption and acceptance of the system.

However, both user feedback and questionnaire responses highlight a limitation of our approach in terms of layout stability—a common challenge for continuous update strategies. Nonetheless, some participants noted that while the layout stability of our method is slightly weaker, it remains within an acceptable range. Furthermore, as we have so far only tested our method within a controlled VR environment, future development must address the complexities of real-world AR scenarios. These scenarios present significantly more complex challenges, including greater environmental variability, more unpredictable user movements, and a wider range of potential occlusions. Therefore, future work adapting our method to these more complex real-world settings should focus on robustifying the algorithm against these factors. This includes developing techniques to maintain label clarity and stability even in the presence of significant noise and occlusion. Additionally, we plan to develop an adaptive parameter adjustment mechanism to streamline the parameter tuning process.

In summary, we introduce a binocular 3D dynamic label layout method leveraging a continuous update strategy. Our approach improves task completion time, particularly in dense labeling scenarios, while simultaneously reducing user workload (physical and temporal demands, and effort). User feedback confirms that our method excels in layout clarity, aesthetics, adaptability, and overall satisfaction. While layout stability presents a minor limitation, it remains within acceptable parameters. Overall, our method offers a solution for dynamic 3D labeling, especially where high label density and user experience are paramount.

7. Conclusions

In this paper, we demonstrate that double vision between labels caused by binocular disparity negatively impacts user experience and introduce a binocular optimization strategy to address this issue. Building on this, we propose a 3D dynamic label layout method based on a continuous updating strategy that incorporates a binocular optimization strategy. This method simultaneously optimizes label layouts from both the left- and right-eye perspectives while applying force-based adjustments to dynamically refine the layout.

Experimental results show that our method effectively reduces the occurrence of double vision, label overlap, and leader line crossings in 3D scenes. Subjective and objective evaluations confirmed that our method significantly improves layout clarity, aesthetics, adaptability, and overall user satisfaction compared to baseline methods. Users were able

to comprehend scene information more efficiently, while experiencing reduced workload, particularly in high-density labeling environments.

This work offers a solution for dynamic 3D labeling, particularly where high label density, binocular viewing, and user experience are important. Future research will focus on extending the applicability of our method to more complex real-world AR scenarios and developing adaptive parameter adjustment mechanisms to further enhance its robustness and usability.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/electronics14112199/s1>, The supplementary materials contain label layout results under different scenes and viewpoints.

Author Contributions: Conceptualization, Z.Z. and Y.H.; Methodology, Z.Z. and Y.H.; Software, Z.Z.; Investigation, Z.Z. and X.X.; Formal Analysis, Z.Z. and X.X.; Visualization, Z.Z. and X.X.; Writing—Original Draft, Z.Z.; Writing—Review and Editing, Y.H.; Supervision, G.Y. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences, grant number No.T403461.

Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki and approved by the Institutional Review Board of University of Chinese Academy of Sciences (protocol code UCASSTEC25-019). Date of approval: 14 April 2025.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The data presented in this study are available on request from the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

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